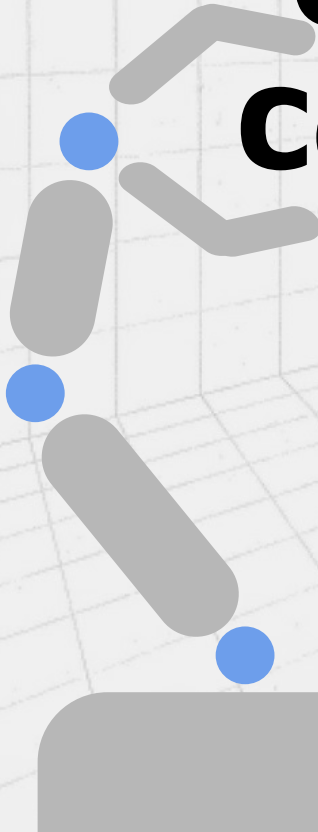
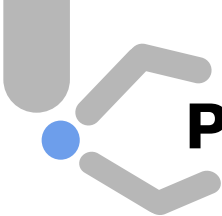


A stylized graphic of a robot arm on the left side of the slide. It consists of several gray rectangular segments connected by blue circular joints. The arm is positioned vertically, with its base at the bottom left and its end reaching towards the top left, partially overlapping the main title.

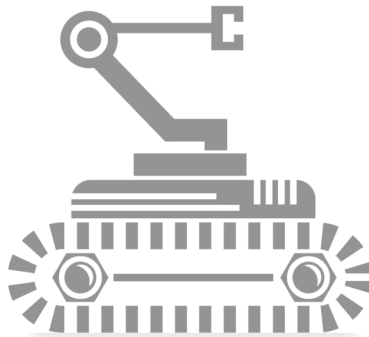
Optimization of Robot Controlling in Industrial Environment

Robocom & FAcom meeting



Problem breakdown

- Data mining
- Multiple robots control
- Supply chain management





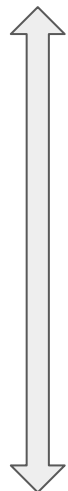
Traditional Approach

AGV/AMR navigation methods

AGV/AGC: Automated Guided Vehicles

AMR: Autonomous Mobile Robots

AGV

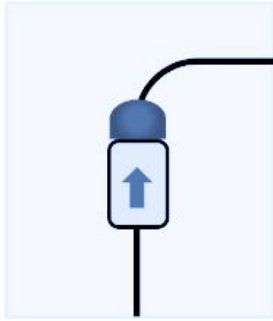


AMR

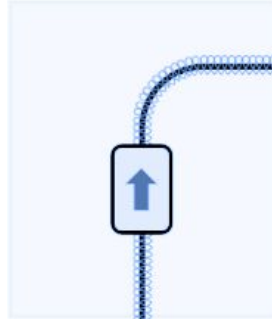
- Wire/Tape navigation
- Magnetic navigation
- Optical Navigation
- Laser-guided navigation (LGV)
- Natural feature-based navigation
 - LiDAR
 - Vision

- Logic implementation is easier
 - Precise movements
 - Not flexible
 - Higher preparation cost
-
- Higher flexibility (environment)
 - Lower preparation cost
 - Complicated logics (high computational cost)

AGV/AMR navigation methods



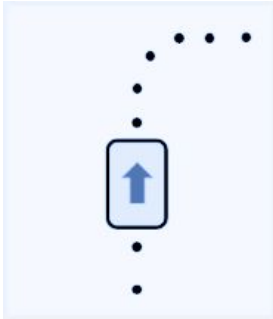
Optical guidance



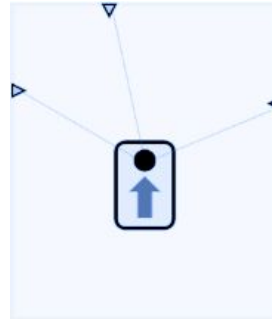
Magnetic guidance



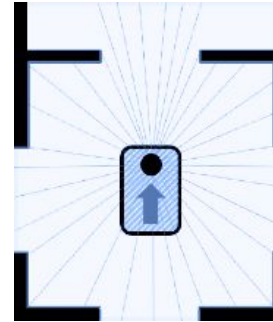
Cartesian reference



Inertial guidance

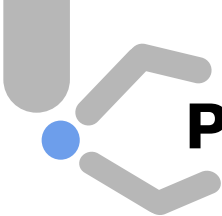


Laser guidance

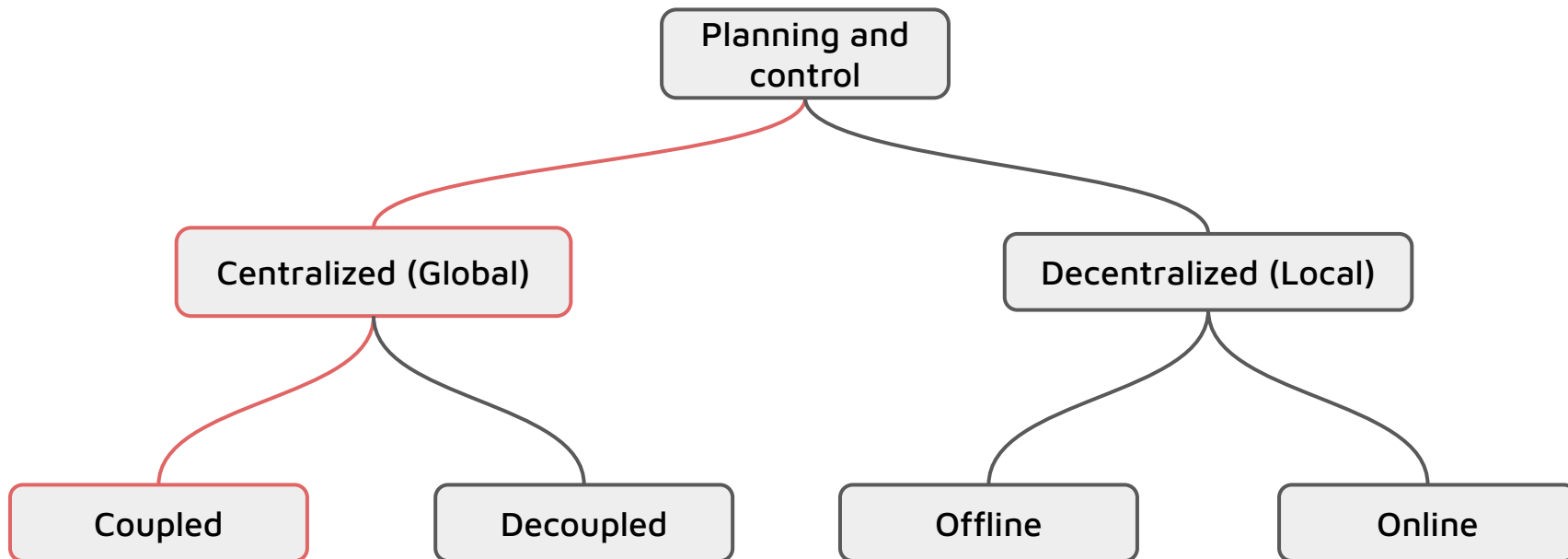


Vision/Laser-based guidance

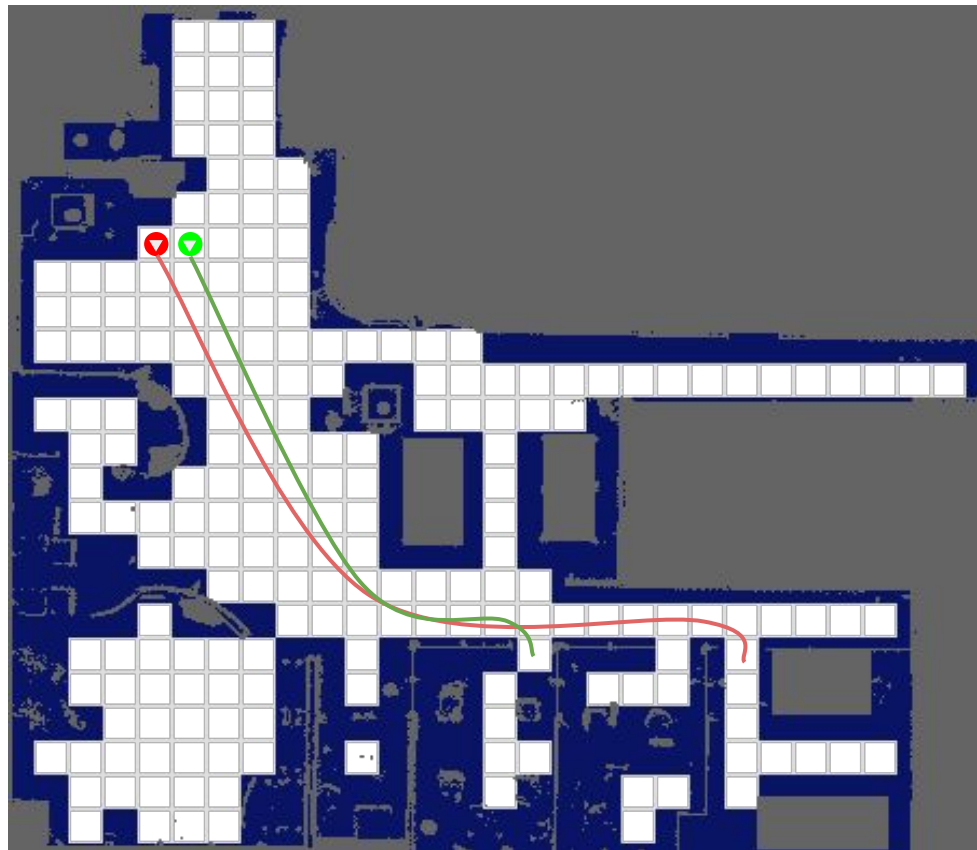
Source: <https://doi.org/10.1016/j.ejor.2021.01.019>



Planning and control of autonomous mobile robots (categorization)



Multi-robot path planning system (Last year)





Specification of the scenario

- Task specific
- Semi-dynamic environment
- Short distance routes (cell to cell movements)
- Centralized control system
- Less human interaction



Factors for optimization

- Moving speed (depends on the environment, max energy consumption, etc.)
- **Maneuverability (mecanum wheel robots)**
- Effective obstacle avoidance methods
- Guiding method
- Multi robot
 - **Scheduling**
 - **Path planning (routing)**

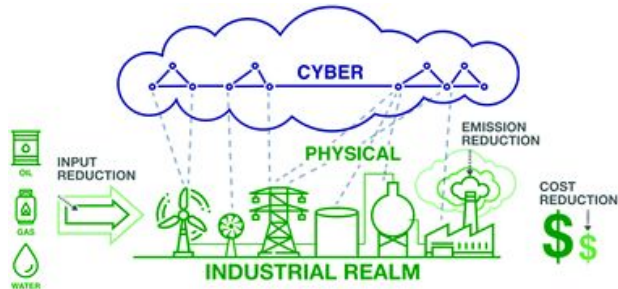


Smart Industry

Industry 4.0

CPS and DT

- Cyber physical system (CPS): Physical and engineered systems whose operations are monitored, coordinated, controlled and integrated by a computing and communication core. (Chen et al.)
- Digital Twin (DT): A virtual representation of a physical object or a process using real data to enable understanding, learning and reasoning.

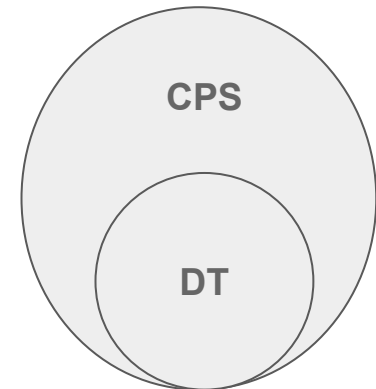


Source: <https://doi.org/10.1039/C9EE01919G>



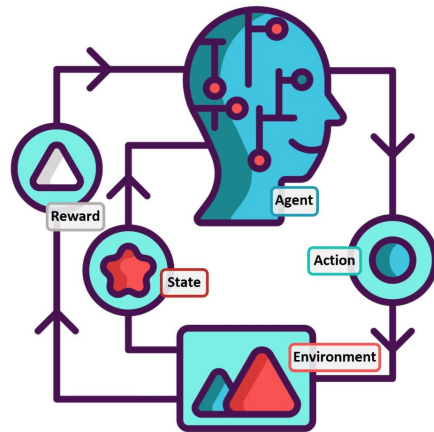
Source:

<https://www.iien.eu/article/leveraging-digital-twin-technology-to-improve-innovation/>



AGV/AMR with reinforcement learning

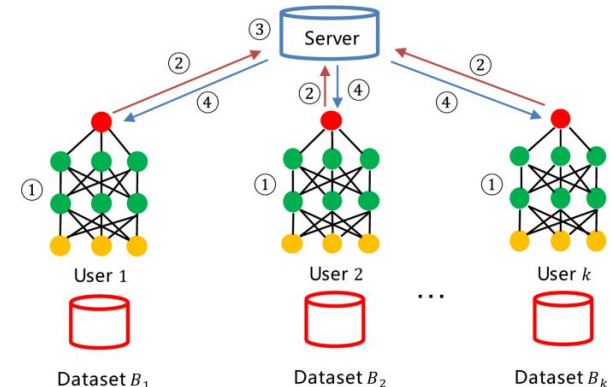
- Reinforcement learning (RL) is an online learning method that takes actions in an environment to maximize cumulative reward.
- RL can be used for both AGVs and AMRs
- Difficult to train a RL agent in a real environment
- RL agents for AGV/AMR can be trained effectively in DT environments.



Source:
<https://towardsdatascience.com/reinforcement-learning-fda8ff535bb6>

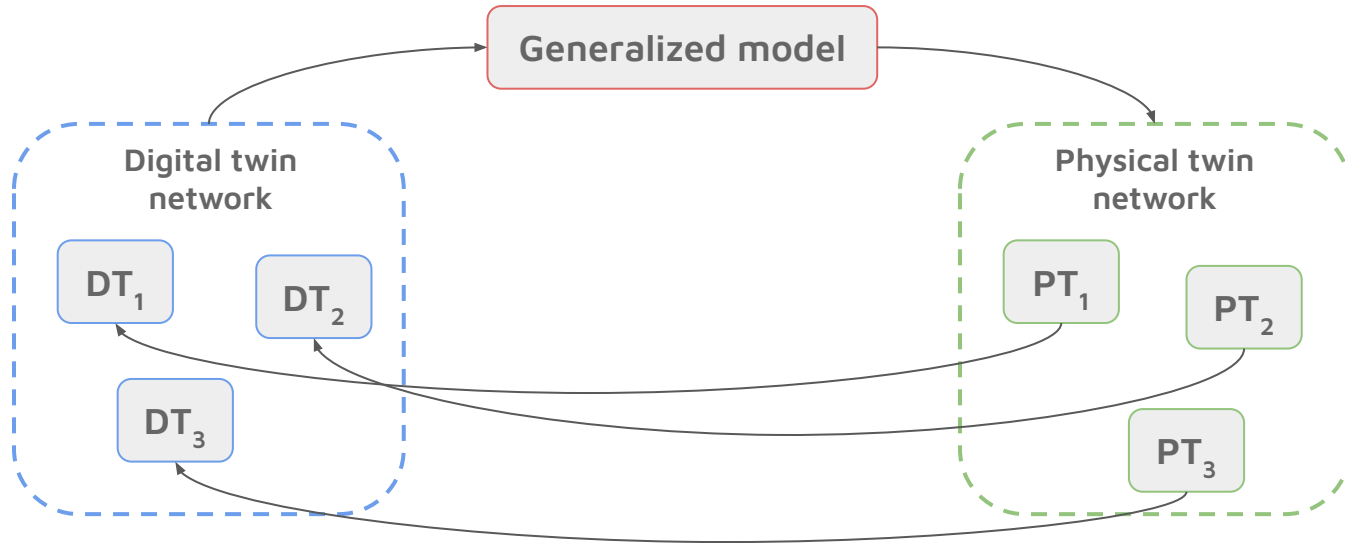
Federated learning (FL)

- Introduced by Google.
- Machine learning technique that trains an algorithm across multiple decentralized edge devices or servers holding local data samples, without exchanging them
- FL can be used to train more generalized RL agent to control AGV/AMR.



Possible approach

- Individual RL agent can be trained in the corresponding DT.
- Trained model data then send to the centralized server to aggregate into generalized model
- Generalized model s cab used for the PTs





References

- Fragapane G, de Koster R, Sgarbossa F, Strandhagen JO. Planning and control of autonomous mobile robots for intralogistics: Literature review and research agenda. European Journal of Operational Research. 2021 Jan 17. DOI: <https://doi.org/10.1016/j.ejor.2021.01.019>
- Chen H. Applications of cyber-physical system: a literature review. Journal of Industrial Integration and Management. 2017 Sep 17;2(03):1750012.
- Rasheed A, San O, Kvamsdal T. Digital twin: Values, challenges and enablers from a modeling perspective. Ieee Access. 2020 Jan 28;8:21980-2012.
- Sun W, Lei S, Wang L, Liu Z, Zhang Y. Adaptive federated learning and digital twin for industrial internet of things. IEEE Transactions on Industrial Informatics. 2020 Oct 29;17(8):5605-14.

Thank you

